

Token Optimizer

Convert files to token-efficient markdown for LLM context

A modern, browser-based tool designed to convert PDFs, Word documents, spreadsheets, code, JSON, data structures, and images into clean, LLM-optimized markdown. **All conversion happens client-side in the browser** — ensuring 100% data privacy since your files never leave your machine.

⚡ Features

- 📄 **Universal File Conversion** — Converts PDFs, .docx, .xlsx, HTML, JSON, CSV, vector/raster images, and plain text to Markdown.
- 📄 **100% Client-Side Processing** — No servers, no APIs, no third-party uploads. Your data remains secure on your device.
- 📄 **Token & Size Optimization** — Strips binary bloat, metadata, and CSS styles, resulting in up to 90%+ size reduction and significant savings on LLM prompt costs.
- 📄 **Tiered Access & Authentication** —
 - Guest Mode: Perform up to 5 conversions with instant client-side local tracking.
 - Supabase Integration: Create a free account via passwordless Email OTP and secure password confirmation to unlock unlimited, forever-free conversions.
- 📄 **Real-time Analytics** — Instantly estimates tokens, compares raw file size against output markdown size, and calculates exact reduction percentages.
- 📄 **Interactive Preview** — Toggle dynamically between rich rendered markdown style sheets and raw markdown source.
- 📄 **Direct Actions** — One-click copy to clipboard or immediate download as a .md file.

- **🔒 Safety Features** — Integrated file size warning systems to prevent browser-memory blockages during heavy conversion pipelines.
- **🌙 Adaptive Theme** — High-fidelity dark mode matching the aesthetic of modern LLM interfaces like Anthropic's Claude.

🔗 Supported Formats

Category	Supported Formats	Engine/Library
Documents	PDF, DOCX, DOC, ODT, RTF	pdfjs-dist & mammoth
Spreadsheets	XLSX, XLS, CSV, TSV, ODS	SheetJS (xlsx)
Presentations	PPTX, PPT, ODP	Native text parser
Web & Markup	HTML, HTM, XHTML	Turndown
Data	JSON, XML, YAML, YML, TOML	Native serializing formatters
Images (OCR/Meta)	PNG, JPG, JPEG, GIF, SVG, WEBP, BMP, ICO	Custom local file rendering
Code	.js, .jsx, .ts, .tsx, .py, .rb, .java, .cpp, .go, .rs, .php, .sql, etc.	Syntax-highlighted codeblocks

🔗 Architecture

File Structure & Roles

- **src/App.jsx** — Root coordinator. Manages global states (theme, active files, visibility of modals) and ties state transitions between the conversion hooks and UI wrappers.
- **src/hooks/useFileConversion.js** — The processing core. Uses lazy dynamic imports to load heavy libraries (like pdfjs-dist or xlsx) only when a user uploads that specific file type. Uses AbortController for clean conversion cancellations.
- **src/hooks/useUsage.js** — Local storage usage tracker. Enforces the guest conversion threshold (up to 5) and bridges session state.
- **src/components/AuthModal.jsx** — Handles the Supabase auth state machine. Supports signup, standard OTP checks, multiple token verification triggers (email, signup, magiclink), new user password initialization, and secure credentials login.
- **src/components/FileSizeWarning.jsx** — Visual alerts when files exceed target size limits to prevent browser crashes.
- **src/components/ProcessingProgress.jsx** — Shows progress indicators for multi-phase operations.
- **src/lib/supabase.js** — Connects to the Supabase backend utilizing env configurations.
- **src/utils/fileUtils.js** — Utility helpers for parsing raw text, estimating Claude-friendly LLM token lengths, and formatting file sizes.

🔗 Getting Started

1. Prerequisites

- Node.js v18.0.0 or higher
- npm v9.0.0 or higher

2. Installation

Clone the repository:

```
git clone https://github.com/Mirza-Glitch/markitdown-js.git
```

Install dependencies:

```
npm install
```

3. Environment Variables

Create a file named .env in the root folder of the project:

```
VITE_SUPABASE_URL=https://your-supabase-url.supabase.co
```

```
VITE_SUPABASE_ANON_KEY=your-supabase-anon-key-here
```

4. Setup Custom SMTP / Resend in Supabase

1. Go to Supabase Dashboard -> Project Settings -> Auth -> Email Settings.
2. Turn Enable Custom SMTP to ON.
3. Configure your SMTP provider (e.g., Resend).
4. Ensure the domain matches perfectly.

5. Setup Google & GitHub OAuth in Supabase

Frontend buttons are configured in AuthModal.jsx to trigger authentication client-side. Configure callback URLs in GitHub and Google Developer Consoles as described in the specifications.

6. Running the Application

Launch the development server:

```
npm run dev
```

Open <http://localhost:5173> in your browser.

7. Building for Production

To bundle the optimized assets:

```
npm run build
```

❓ Why Optimize Tokens?

Large language models process text in units called tokens. Feeding raw, formatted files (like .pdf or .docx) into a chat window includes a massive volume of invisible XML metadata, styling markup, and binary overhead.

Optimizing your context:

- 1. Reduces API Billing: Smaller prompt packages mean you pay less per API query.*
- 2. Preserves Context Limits: Fit more actual information inside the context window.*
- 3. Enhances Reasoning Quality: Eliminates distracting noise, letting the model focus solely on clean text structures.*

❓ License

This project is for personal and educational use.